

Lecture 8 — α , the Honest Chapter

Keeper, for Casey Koons and the team

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The question

Is 137 in the geometry?

Every reader of this program asks it first, and it is the right question, because the fine-structure constant $\alpha \approx 1/137.036$ is the number a theory of this kind is *for*. It is dimensionless, it is measured to eleven digits, and the Standard Model takes it as an input. A geometry that produced it would be doing something no theory has done. This lecture tells you what we found, what we proved cannot work, and whose thread we are holding while we look. It is the chapter we most wanted to write differently.

For the reader in a hurry

Our shape has a few whole numbers built into it, and one simple combination of them — three cubed, times five, plus two — is 137. That is striking, and for a while we thought it was the answer. Then we checked how striking it really is, by asking how many *other* famous numbers a shape like ours could produce the same way. The honest answer was: quite a lot. So the coincidence is a clue, not a proof. We then tried the one route that *would* have been a proof — reading α off the shape's volume, the way a physicist named Wyler tried in 1969 — and computed the answer with the target hidden from us. It came out $8\pi^3/3 \approx 82.7$, not 137. That route is closed, for good, and we say so in the same voice we would have used to announce success.

What is true

Three things about 137 are simply true and nobody disputes them.

1. $\alpha^{-1} = 137.035999 \dots$ is measured, to about one part in 10^{10} .
2. $N_{\max} = N_C^3 n_C + \text{rank} = 27 \cdot 5 + 2 = 137$ is a computed invariant of the shape — an integer combination of numbers that were fixed before anyone compared it with anything (T186; registry, March 2026). It also has an honest meaning

inside the program: it is the number of independent boundary channels, a capacity, and it enters the definition of the program's clock (Lecture 4).

3. The two numbers agree to three significant figures and disagree at the fourth.

Everything else in this lecture is about the word that goes between them.

Why the forward form is a mnemonic and not evidence

In August we did what our own reviewer paper ("Forcing and Evidence") tells everyone else to do: we ran a null. Take the small integers the shape provides, form every expression of the allowed kind, and count how many measured constants of nature land within the same tolerance. If 137 is special, few targets should score as well. It is not special. Twenty-three to twenty-six of seventy-six targets scored higher (K1813); the real attractors of the integer soup are $128 = 2^8$ and $125 = n_C^3$, and 137 sits in the bulk. For comparison, the same instrument on the same day gave the program's muon-mass form fewer than one chance in sixty of being an accident (K1814). The instrument works. On the famous number it says: coincidence.

So $\alpha^{-1} = N_C^3 n_C + \text{rank}$ is identified. It is a good way to remember the number. It is not a reason to believe the shape produces it, and from 2026-08-11 the registry has carried the instruction, in those words, not to cite it externally as derived. This document's own scorecard failed to obey that instruction for twelve days (K1816); we record the failure so you can weigh the rest.

The route that would have been a proof, and how it failed

Armand Wyler, in 1969 and 1971, noticed that a ratio of volumes of bounded symmetric domains — the very family our object belongs to — comes remarkably close to α . His formula was criticised the same year by B. Robertson in *Physical Review Letters* — the volume ratio matches only if the radius of the domains is arbitrarily set to one, and one coefficient of a Poisson kernel was wrong — and the word that stuck to it afterwards was numerology, for fifty years. Whether or not it was fair, it named the right worry: a volume ratio has many readings, and if you may choose the reading after seeing the target, you can hit anything.

We are the natural heirs to that idea, and we tried to do it properly. In August 2026 we asked: is there a *forced* reading — one the geometry picks out with no choice left to us — of a volume on D_{IV}^5 that should equal α^{-1} ? There is. The bare vertex on the Šilov boundary, with its power fixed by the Born rule and the family of candidate landings and their tolerances written down and hashed *before* the computation, gives one number. We computed it half-blind (K1826, toy 5496):

$$\alpha_{\text{forced vertex}}^{-1} = \frac{8\pi^3}{3} \approx 82.68,$$

the Hua volume of the Lie sphere. Not 137. Under the pre-registered protocol this is Landing C — the miss — and it closes not just one formula but the class: the forced

member is refuted by computation, and the free family (every reading that leaves a choice to the reader) had already been shown by a separate null to fit anything. Two legs, one pincer. The volume reading of α on D_{IV}^5 is dead, and it died the way a scientific claim should — with the target hidden and the failure certified.

Wyler had a correct instinct and no way to kill his own idea; his critics had a correct worry and no way to test it. We had both, and used them on ourselves.

Two more identifications, named as such

The electron’s anomalous moment. A 2026 paper of ours expressed the two-loop QED coefficient of a_e — written C_2 in the QED literature, and called A_2 here so that it is not confused with the program’s integer $C_2 = 6$ of Lecture 1 — as a sum of four terms from a trace on the geometry. In September we compared the four terms with the literature (K1872). They are, term for term, the four summands of Petermann and Sommerfield’s 1957 closed form,

$$A_2 = \frac{197}{144} + \frac{\pi^2}{12} - \frac{\pi^2}{2} \ln 2 + \frac{3}{4} \zeta(3) = -0.32848 \dots$$

The match is exact by identity — term for term — not an independent numerical agreement. The derivation is theirs. Our row is re-tiered to *identified*, the word “crown jewel” is retired from the legacy chapter that used it, and the result is still interesting for a reason we can state: a trace on this geometry reproduces the *structure* of the QED coefficient, and why it should is an open question, not a solved one.

Asymptotic freedom. The one-loop coefficient of the strong coupling’s running, $b_0 = 11 - \frac{2}{3}n_f$, equals 7 at six flavours. The program’s integer $g = 7$ was read as this coefficient. It is an identification (K1875): the coefficient is matched, not produced — 7 at six flavours, 23/3 at five, so the reading is flavour-dependent — and on the *sign*, two register entries disagree: T2526 (re-scoped 2026-08-22) carries the sign as derived *given the imported SU(3) and $N_c = 3$* , while K1875 says sign and coefficient are textbook and not derived from D_{IV}^5 . Until that collision is ruled (Keeper’s, owed), the sign is quoted with T2526’s scope clause and nothing stronger.

And a small one. $\ln 137$ appeared in several of our expressions at the level of the 137th channel. It is a normalisation unit — a choice of where to set a scale — and not a quantity the geometry predicts (K1873).

A withdrawal

In September one of us wrote that the program predicts a slow drift in α , and named laboratory bounds it might fail against. That was wrong, and it was caught within the day. The program’s clock (T1136) *defines* its α through the success of a write — it is a ratio fixed by the definition — so it is constant by construction, and no laboratory drift bound reaches anything the program owns. We withdrew the claim

(K1884). A falsifier you cannot fail is not a falsifier, and claiming one is the same error as claiming a derivation.

Whose thread we are holding

Wyler's. He asked whether α lives in the geometry of a bounded symmetric domain, and he was not wrong to ask. What he lacked was a pre-registered negative of his own: a computation with the target hidden that says *this* reading does not work, so that the next person does not try it again. We have supplied one for the forced reading on this domain, and the instrument that shows the forward form is in the bulk. (Whether the 1971 critiques already amounted to such a negative is a question for the literature pinned below, and we do not claim priority over it.) That is what we have added to his program. It is smaller than we hoped. Whether the 1971 critiques already amount to a certified negative on the volume reading is a question for the literature we have just pinned, not a claim we make.

The open question is exactly the one he left: if α is in this geometry, it is not in its volume. Where, then? Every rate mechanism offered so far is dead — push and write are one event, so their ratio is one, and no rate in the dictionary is of order $1/137$ (T1136's annotation of 2026-09-09; K1888). The clock *defines* its α through the success of a write and does not derive it. The open item is the hole, stated as one: nothing in the dictionary carries α . That is where a future lecture would begin.

Tier line

- Measured: $\alpha^{-1} = 137.036$.
- Computed invariant: $N_{\max} = 137$, a capacity of the boundary.
- Identified — do not cite externally as derived: every expression for α^{-1} , including the forward form; the a_e coefficient; $b_0 = 7$.
- Fired and lost, certified: the volume reading — forced vertex $= 8\pi^3/3$.
- Withdrawn: α drift.
- Open: whether α is in the geometry at all.

What would make this lecture wrong

Nothing here is exposed to a laboratory number; that is the point. The lecture would become *wrong in the other direction* — under-claiming — if someone derives α from the geometry by a route that is not a volume reading and not a forward form, with the target hidden and the protocol pre-registered. We would be delighted. The instruments to check it are in the repository.

Where to look

K1813, K1814 and the “Forcing and Evidence” paper for the null; K1826 with toy 5496 and its pre-registration hash for the forced vertex; K1872 for the term-by-term comparison with Petermann–Sommerfeld; K1873, K1875; T1136 and K1884 for

the drift withdrawal; A. Wyler, “L’espace symétrique du groupe des équations de Maxwell,” *C. R. Acad. Sci. Paris* 269 (1969) A743–A745; A. Wyler, “Les groupes des potentiels de Coulomb et de Yukawa,” *C. R. Acad. Sci. Paris Sér. A* 272 (1971) 186–188 (11 January 1971; pinned 2026-09-14 from the OSTI catalogue record of the Lawrence Radiation Laboratory translation UCRL-TRANS-10557, OSTI ID 4011633, “Compt. Rend., Ser. A 272: 186-8” — MathWorld’s “271” is wrong; volume 271 is the second semester of 1970); B. Robertson, “Wyler’s Expression for the Fine-Structure Constant α ,” *Phys. Rev. Lett.* 27, no. 22 (29 Nov 1971) 1545–1547, doi:10.1103/PhysRevLett.27.1545 (pinned 2026-09-11 from the OSTI record; the two-part critique — the radius set arbitrarily to one; a wrong Poisson-kernel coefficient for the type IV domain — is as reported by secondary sources and should be read from the paper before it is quoted as Robertson’s words); R. Gilmore, “Scaling of Wyler’s Expression for α ,” *Phys. Rev. Lett.* 28 (1972) 462–464.